SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.3: Temperature and Thermal Expansion — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Temperature and Thermal Expansion
Driving Question	DRIVING QUESTION: Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Encounter and Predict: The Cracked Frozen Bottle**	Teacher presents the anchoring phenomenon: a sealed plastic water bottle left in a Nairobi household freezer overnight has cracked and bulged. Students examine photographs or a physical specimen and record initial noticings. Teacher elicits predictions using a cold-call round (minimum 5 students): 'What do you think happened inside this bottle?' Students then observe a live demonstration — a water-filled balloon placed in a freezer versus one left at room temperature — and record comparative observations about volume and shape change. Pedagogical note: Resist correcting misconceptions at this stage; the goal is to surface prior knowledge and anchor curiosity. Document student predictions on the Driving Question Board (DQB) for revisitation in Lesson 6.	Students are introduced to the phenomenon that water, unlike most substances, expands when it freezes rather than contracting. Teacher explains that this is called anomalous expansion and that it is caused by the hydrogen bonding structure water molecules form upon solidification — a hexagonal lattice that is less densely packed than liquid water. Key vocabulary introduced: thermal expansion, anomalous expansion, freezing point, volume, pressure. Teacher emphasises that most matter contracts upon cooling and expands upon heating, making water's behaviour a scientific puzzle worth investigating. Pedagogical note: Use the particle model (dots on the board) to give students a visual anchor — show closely packed liquid particles versus the more spaced hexagonal ice lattice. This initial model is deliberately incomplete; students will revise it across the unit.	Students begin constructing their Initial Scientific Model: a particle-level sketch showing liquid water molecules in a bottle before and after freezing. Teacher guides the class to annotate the 'after' sketch with arrows showing outward pressure on the bottle walls. DQB is formally started — teacher posts the driving question and invites students to add sub-questions (e.g., 'Does all matter expand when heated?', 'Why does ice float on Lake Victoria?', 'What kept the bottle from stretching?'). Pedagogical note: Validate all questions on the DQB regardless of accuracy; mark them as 'To Investigate' and return to each as the unit progresses. The Initial Model and DQB together serve as the unit's formative baseline — photograph or collect them for comparison in Lesson 6.
Lesson 2 **Temperature Scales and Thermometers:**	Students carry out a structured observation activity using three types of thermometers: a liquid-in-glass thermometer (mercury or	Temperature is formally defined as a measure of the average kinetic energy of particles in a substance — the higher the	Students connect temperature measurement to the anchoring phenomenon: 'The bottle was placed in a

Measuring the 'How Hot'	alcohol), a digital probe thermometer, and a bimetallic strip thermometer (demonstrated by teacher). Each group measures the temperature of: (a) a cup of boiling water (ugali-cooking temperature), (b) tap water at room temperature, and (c) cold water from a cooler box. Students record readings in Celsius from all three instruments and compare results. Teacher then shows a calibration demonstration: both the fixed lower point (ice-water mixture at 0 °C) and upper fixed point (steam at 100 °C at Nairobi's altitude ~1,661 m) are established. Pedagogical note: Nairobi's altitude slightly lowers the boiling point of water (~94 °C); use this as a real-world discussion hook about how altitude affects cooking times for githeri or maize.	temperature, the faster the particles move. Students learn that a thermometric property is any measurable physical property that changes continuously and reliably with temperature (e.g., length of a liquid column, electrical resistance, colour of a bimetallic strip). Key scales covered: Celsius (°C), Kelvin (K), and Fahrenheit (°F), with conversion formulae $T(K) = T(°C) + 273$ and $T(°F) = 1.8 \times T(°C) + 32$. Teacher emphasises that Kelvin is the SI unit and that absolute zero (0 K) represents the theoretical point of zero particle motion. Pedagogical note: Use wait time of at least 8 seconds after asking 'What does temperature actually measure?' before cold-calling — this concept requires think time. Expect the common misconception that temperature measures 'heat'; address it explicitly.	freezer where the temperature dropped below 0 °C. Using the Kelvin scale, what happened to the average kinetic energy of the water molecules as the temperature fell?' Teacher cold-calls 3 students to articulate the particle-energy link. Students update their Scientific Model to show that lower temperature → slower particle motion → molecules settling into a more ordered (hexagonal) structure. DQB Update: Students answer the posted sub-question 'How do we measure temperature accurately?' and add new questions such as 'Does the amount of expansion depend on how much the temperature changes?' and 'Are different materials affected equally?' Pedagogical note: Ensure all groups have correctly converted the freezer temperature (approx. −18 °C) into Kelvin as a formative check before moving on.
Lesson 3 **Linear Expansion of Solids — The Stretching Rod**	Students conduct the metal rod (or rail) linear expansion experiment. Each group heats a measured metal rod (steel or copper) using a Bunsen burner or hot-water bath and measures the change in length using a micrometer screw gauge or a linear expansion apparatus. Data recorded: original length (L_0), temperature change (ΔT), and change in length (ΔL) for at least three different temperature increments. Groups then plot ΔL against $L_0 \Delta T$ on graph paper and observe a straight-line relationship passing through the origin. Pedagogical note: Circulate during data collection and prompt precision — insist on at least three trials per temperature increment and calculation of a mean ΔL. Ask groups: 'What does the gradient of your graph represent?' Allow 6 seconds wait time before taking responses.	From their own experimental gradient, students derive the coefficient of linear expansivity $\alpha = \Delta L / (L_0 \Delta T)$, with SI unit K^{-1} (or $°C^{-1}$). The working formula $\Delta L = \alpha L_0 \Delta T$ is formalised. Students compare their experimentally derived α values for steel (~12×10^{-6} K^{-1}) and copper (~17×10^{-6} K^{-1}) with published values, computing percentage error. Teacher contextualises: engineers at the Nairobi Expressway and Kenya Railways must account for linear expansion when designing rail joints and bridge expansion gaps. If expansion is ignored, rails buckle in the midday Rift Valley heat. Key vocabulary: coefficient of linear expansivity, original length, temperature change, expansion gap, percentage error. Pedagogical note: Emphasise that α is a material constant — it tells us how a specific material responds, which is why different materials behave differently under the same temperature change.	Students apply $\Delta L = \alpha L_0 \Delta T$ to a Kenya-relevant problem: 'A steel rail on the SGR (Standard Gauge Railway) between Nairobi and Mombasa is 25 m long at 20 °C. If the temperature rises to 45 °C on a hot afternoon, by how much does the rail expand?' ($\alpha_{steel} = 12 \times 10^{-6}$ K^{-1}; expected answer: $\Delta L = 7.5$ mm.) Teacher connects this back to the anchoring phenomenon: 'The bottle is a container, just like a rail is a structure — when the contents inside expand, they exert an outward force. What happens when there is no room to expand?' DQB Update: The question 'Does the amount of expansion depend on temperature change?' is marked as ANSWERED with evidence from the graph. New question added: 'What about expansion in two or three dimensions — does it matter for the bottle?' Students annotate their Scientific Model to show particle spacing increasing in one dimension along a heated rod.
Lesson 4 **Superficial and Volumetric Expansion —**	Students observe two demonstrations: (1) The Ball-and-Ring demonstration — a metal ball that fits exactly through a ring at room	Teacher leads a guided mathematical derivation showing that for a solid that expands linearly with coefficient α:	Students now have the conceptual and mathematical tools to address the anchoring phenomenon volumetrically.

From Length to Volume	temperature no longer fits after being heated, illustrating superficial (area) expansion. Teacher asks: 'What would happen if we heated the ring instead? What if we heated both?' Cold-call 4 students and record predictions before revealing the answer. (2) A volumetric expansion simulation or video showing a liquid in a flask expanding into a capillary tube when heated, making volume change directly visible. Students record qualitative observations and estimate the ratio of volume change to area change to length change from the simulation data. Pedagogical note: The ball-and-ring demonstration is high-engagement — use it to drive home the idea that expansion occurs in ALL directions simultaneously, setting up the mathematical derivation that follows.	superficial (area) expansivity $\beta \approx 2\alpha$, and volumetric (cubic) expansivity $\gamma \approx 3\alpha$. The derivation uses the binomial approximation for small expansions, which teacher works through step-by-step on the board. Key formulae: $\Delta A = \beta A_0 \Delta T$ and $\Delta V = \gamma V_0 \Delta T$. Students also learn that liquids only exhibit volumetric expansion (they have no fixed shape) and that γ for liquids is generally much larger than for solids (e.g., $\gamma_{water} \approx 207 \times 10^{-6}\ K^{-1}$ vs. $\gamma_{steel} \approx 36 \times 10^{-6}\ K^{-1}$). Apparent vs. real expansion of liquids in containers is introduced. Key vocabulary: superficial expansivity, volumetric expansivity, apparent expansion, real expansion, binomial approximation. Pedagogical note: Many students struggle with the derivation — provide a worked numeric example immediately after the algebra, using a concrete Kenyan object (e.g., a metal sufuria lid) to ground the abstraction.	Teacher poses: 'The sealed water bottle had a volume of 500 mL at 20 °C. Using $\gamma_{water} = 207 \times 10^{-6}\ K^{-1}$, calculate the expected volume change if the water were cooled to 4 °C — then explain why the actual cracking occurs below 4 °C and involves a different mechanism.' This two-part question distinguishes normal volumetric contraction (cooling from 20 °C to 4 °C) from anomalous expansion (freezing below 4 °C), previewing Lesson 6. DQB Update: 'Does expansion work the same in all directions?' is marked ANSWERED. New questions: 'Why does water behave differently from other liquids near 0 °C?' and 'Could engineers design a bottle that survives freezing?' Students revise their Scientific Model to show expansion in all three spatial dimensions, with annotations for β and γ.
Lesson 5 **Applications and Consequences of Thermal Expansion**	Students investigate real-world applications of thermal expansion through a structured stations activity (or teacher-led gallery walk). Station examples: (A) Photograph and data set of expansion joints on the Nyali Bridge in Mombasa — students measure the gap width and calculate the expected temperature range the bridge must accommodate. (B) A bimetallic strip that bends when heated — students sketch the before/after shape and explain using differential expansion ($\alpha_{brass} > \alpha_{iron}$). (C) A thermostat circuit diagram showing how a bimetallic strip acts as a temperature-controlled switch. (D) A video or diagram of a thermometer being calibrated. (E) A cracked clay pot left near a jiko fire — students explain uneven thermal expansion causing fracture. Students record observations and explanations at each station. Pedagogical note: Stations A and E are most directly connected to the anchoring phenomenon — ensure all groups spend adequate time at these two stations and record explicit connections in	Students consolidate understanding of how thermal expansion is both useful (thermostats, thermometers, bimetallic strip devices, riveting of metal sheets) and potentially damaging (cracking of roads, bursting of pipes, warping of rail tracks, breaking of bottles). Key engineering design principle: materials with different α values can be paired (bimetallic strips) to convert temperature change into mechanical movement. Alternatively, expansion joints and gaps are engineered into structures to allow safe, unconstrained expansion. Students are introduced to the concept of thermal stress — the internal force per unit area generated when a material is prevented from expanding freely. Pedagogical note: Connect the Kenyan context explicitly — ask students why tarmac roads in Nairobi often crack during the long rains (thermal cycling + moisture expansion), and why SGR rail joints are spaced more widely in the hotter coastal section near Mombasa than the cooler highland section near Nairobi.	Students write a structured explanation paragraph (using the Claim–Evidence–Reasoning framework): 'A water pipe running along an outdoor wall in Eldoret cracked overnight. Using your knowledge of thermal expansion, explain what happened and propose one design change that could prevent this.' Teacher circulates and provides targeted feedback. DQB Update: Sub-questions about engineering applications ('How do engineers use thermal expansion safely?') are marked ANSWERED with evidence from stations. Students add a final preparation question: 'Can we now fully explain the cracked frozen bottle — including the role of hydrogen bonding?' Students annotate their Scientific Model to include bimetallic strip behaviour and thermal stress arrows, indicating what happens when expansion is constrained.

	their Summary Table.		
Lesson 6 **Model Revision, Synthesis, and DQB Resolution — Explaining the Crack**	Teacher facilitates a full-class Model Revision Gallery: each group posts their evolving Scientific Model (from Lesson 1 through Lesson 5) on the wall. Groups do a silent gallery walk, using sticky notes to mark: (1) one strength of each group's model, and (2) one gap or question. Teacher then presents the final evidence set: a molecular animation or diagram showing hydrogen bonding in liquid water versus the hexagonal ice lattice, paired with a density-vs-temperature graph for water (showing the density maximum at 4 °C and the sharp density drop upon freezing). Students observe that ice is ~9% less dense than liquid water, directly explaining the ~9% volume increase that ruptures the sealed bottle. Pedagogical note: Allow at least 10 minutes for the gallery walk — this metacognitive reflection is essential. Cold-call students to articulate what changed in their model between Lesson 1 and Lesson 6.	Students consolidate the complete scientific explanation: (1) Hydrogen bonds between water molecules (H_2O) cause them to arrange into a hexagonal, open lattice upon freezing — this is anomalous expansion. (2) The volume of ice is approximately 9% greater than the same mass of liquid water. (3) When water freezes inside a sealed, rigid container (the bottle), the expanding ice exerts enormous outward pressure — sufficient to crack plastic or glass. (4) This is distinct from normal thermal expansion ($\gamma = 3\alpha$) because it is driven by a phase change and hydrogen bonding, not merely increased kinetic energy. Key vocabulary reviewed: hydrogen bonding, hexagonal lattice, anomalous expansion, density, phase change, thermal stress, volumetric expansivity. Pedagogical note: Emphasise the three-dimensional learning synthesis — students have used SEPs (modelling, data analysis, explanation), DCIs (particle nature of matter, thermal expansion), and CCCs (patterns, cause and effect, structure and function) across all six lessons.	Students write their Final Explanation for the driving question — individually, in full sentences, referencing particle diagrams, data from experiments, and formulae used in the unit. Sentence starters provided: 'The bottle cracked because... / At the particle level, this happened because... / The key difference between water and most other substances is... / Engineers can prevent this by...' DQB Resolution: Teacher leads the class through every question posted on the DQB since Lesson 1, marking each as ANSWERED or STILL INVESTIGATING, and facilitating a brief discussion for any unresolved questions. Final Scientific Model is completed — it must include: particle-level representation of solid/liquid/gas, the anomalous expansion of water, the formulae $\Delta L = \alpha L_0 \Delta T$ and $\Delta V = \gamma V_0 \Delta T$, and at least one engineering application. Pedagogical note: Collect Final Explanations as a summative formative assessment. Use a 3-point rubric: (1) correct particle-level cause, (2) correct formula application, (3) connection to engineering or real-world consequence. These provide direct evidence of readiness for the end-of-strand assessment.